

Blocco 3

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Esercizio 1. Let G be any countable group. Consider $2^G, (2^\omega)^G$ equipped with the shift action. Prove that there exists an injective, G -equivariant function from 2^G to $(2^\omega)^G$.

Proof. Consider $f : 2^G \rightarrow (2^\omega)^G$, $x \mapsto x^*$, where $\forall g \in G, n < \omega$ $x^*(g)(n) := x(g)$. We want to prove:

- f is G -equivariant:

Let $g \in G, x \in 2^G$. We want to prove that $f(g \cdot x) = g \cdot f(x)$. We evaluate both and check they are the same:

$$f(g \cdot x)(h)(n) = (g \cdot x)^*(h)(n) = (g \cdot x)(h) = x(g^{-1} \cdot h)$$

$$(g \cdot f(x))(h)(n) = (g \cdot x^*)(h)(n) = x^*(g^{-1} \cdot h)(n) = x(g^{-1} \cdot h)$$

- f is injective:

Let $x_1 \neq x_2 \in 2^G$. Then $x_1(g) \neq x_2(g)$ for some $g \in G$. Then $x_1^*(g)(0) = x_1(g) \neq x_2(g) = x_2^*(g)(0)$. Thus $x_1^* \neq x_2^*$.

□

Esercizio 2. Let G, H be countable groups and Y a Polish space. Prove that if $\{1\} \neq G \leq H$, then:

1. $E(G, Y) \leq_B E(H, Y)$
2. $F(G, Y) \leq_B F(H, Y)$

Proof. Fix $y_0 \in Y$. Then consider $f : Y^G \rightarrow Y^H$, $p \mapsto p^*$, where $p^*(h) := \begin{cases} p(h) & \text{if } h \in G \\ y_0 & \text{else} \end{cases}$

We must prove:

- f is a reduction from $E(G, Y)$ to $E(H, Y)$

$$- \forall p, q \in Y^G, g \in G \left(p = g \cdot q \Rightarrow p^* = g \cdot q^* \right)$$

Let's consider two cases:

$$* \ h \in G$$

$$p^*(h) = p(h) = (g \cdot q)(h) = q(g^{-1} \cdot h) = q^*(g^{-1} \cdot h) = (g \cdot q^*)(h)$$

* $h \notin G$

$$p^*(h) = y_0 = q^*(g^{-1} \cdot h) = (g \cdot q^*)(h)$$

$$- \forall p, q \in Y^G, h \in H \left(p^* = h \cdot q^* \Rightarrow (\exists g \in G p^* = g \cdot q^*) \right)$$

Let $h \in H \setminus G$ such that $p^* = h \cdot q^*$. Then

$$\forall t \in H \quad p^*(t) = (h \cdot q^*)(t) = q^*(h^{-1} \cdot t)$$

In particular

$$\forall g \in G \quad y_0 = p^*(h \cdot g) = q^*(g) = q(g)$$

Then $q = \text{cost}_{y_0}$. Then $q^* = \text{cost}_{y_0}$. Then $p^* = \text{cost}_{y_0}$. Then $p^* = 1_H \cdot q^*$.

$$- \forall p, q \in Y^G, g \in G \left(p^* = g \cdot q^* \Rightarrow p = g \cdot q \right)$$

$$\begin{aligned} p^* = g \cdot q^* &\Rightarrow \forall h \in H \quad p^*(h) = q^*(g^{-1} \cdot h) \\ &\Rightarrow \forall t \in G \quad p(t) = p^*(t) = q^*(g^{-1} \cdot t) = q(g^{-1} \cdot t) = (g \cdot q)(t) \end{aligned}$$

- f is Borel

$$(s, r) \in f \iff \left((\forall g \in G \quad s(g) = r(g)) \wedge (\forall h \in H \setminus G \quad r(h) = y_0) \right)$$

For the second part, consider $f' : ((Y))^G \rightarrow ((Y))^H$, $p \mapsto p^*$, where $p^*(h) := \begin{cases} p(h) & \text{if } h \in G \\ y_0 & \text{else} \end{cases}$

We claim as above that this is the desired reduction.

- well defined:

We have to check that $\forall p \in ((Y))^G \quad p^* \in ((Y))^H$. Let $h \in H$ such that $h \cdot p^* = p^*$. Then $\forall t \in H \quad (h \cdot p^*)(t) = p^*(t)$. Equivalently $\forall t \in H \quad p^*(h^{-1} \cdot t) = p^*(t)$. In particular $\forall g \in G \quad p^*(h^{-1} \cdot h \cdot g) = p^*(h \cdot g)$. Then $\forall g \in G \quad p^*(g) = p^*(h \cdot g)$. Now there are two possibilities:

- $h \in H \setminus G$:

In this case we have $\forall g \in G \quad p(g) = p^*(g) = p^*(h \cdot g) = y_0$. This means that $p = \text{cost}_{y_0}$. Since $G \neq \{1\}$ we have $p \notin ((Y))^G$, a contradiction.

- $h \in G$:

In this case we have $\forall g \in G \quad p(g) = p^*(g) = p^*(h \cdot g) = p(h \cdot g) = (h^{-1} \cdot p)(g)$. This means that $p = h^{-1} \cdot p$. Since $p \in ((Y))^G$, it must be $h^{-1} = 1$. Then $h = 1$, as desired.

- reduction:

We must prove that $\forall p, q \in ((Y))^G \quad \left(pF(G, Y)q \iff p^*F(H, Y)q^* \right)$. This follows basically from f being a reduction. In fact: $pF(G, Y)q \iff pE(G, Y)q \iff f(p)E(H, Y)f(q) \iff f'(p)E(H, Y)f'(q) \iff f'(p)F(H, Y)f'(q)$

- Borel:

This follows from the Borelness of f , since $f' = f \cap \left(((Y))^G \times ((Y))^H \right)$

□

Esercizio 3. Let E, F cber on X . Prove that if $E \subseteq F$ and F is treeable, then E is also treeable.

Proof. Let T be a Borel graphing witnessing the treeability of F . Let $F = E_G^X$ for some countable group G . Let's fix an enumeration: $G = \{g_n : n < \omega\}$. Call $\langle x_0, \dots, x_n \rangle \in X^{<\omega}$ a T -path if $\forall i < n \ x_i T x_{i+1}$. Given $x \in X$ and an E -class C , denote $PATH_{x,C} := \{s \in X^{<\omega} : s \text{ is a } T\text{-path starting from } x \text{ and ending in } C\}$. Now, given x, C as above we can define $\phi : PATH_{x,C} \rightarrow \omega^{<\omega}$, $\langle x, y_1, \dots, y_k \rangle \mapsto \langle i_1, \dots, i_k \rangle$, where $i_j = \min\{n \in \omega : g_n \cdot y_{j-1} = y_j\}$ (consider x as y_0 to evaluate i_1). In particular $\phi(x) = \emptyset$. One can consider the lexicographic order on $\omega^{<\omega}$, $\leq_{lex}$, where in particular we put $s \leq t$ if s is an initial segment of t . Since ϕ is injective and $\leq_{lex}$ is a well order, we can define a well order on $PATH_{x,C}$, $\leq_{x,C}$, by setting

$$\forall s, t \in PATH_{x,C} \ (s \leq_{x,C} t \iff \phi(s) \leq_{lex} \phi(t))$$

Now consider $\rho : X^2 \rightarrow X$, with

$$\rho(x, y) := \begin{cases} \text{the last element of } \min(PATH_{x,[y]_E}) & \text{if } x F y \\ x & \text{else} \end{cases}$$

Note that $\rho(x, y) = x \iff (x E y \vee x \not F y)$. Now we can define a Borel treeing for E . Consider $t \subseteq X^2$ given by

$$x_1 t x_2 \iff \left(x_1 E x_2 \wedge (\exists i < \omega \exists j < \omega (g_i \cdot x_1 T g_j \cdot x_2 \wedge \rho(g_i \cdot x_1, x_1) = x_1 \wedge \rho(g_j \cdot x_2, x_2) = x_2)) \right)$$

We have to prove the following claims to conclude.

- t is Borel

This follows from the Borelness of E, T, ρ . we prove the Borelness of ρ below.

- ρ is Borel

$$(x, y, z) \in \rho \iff \left((x = z \wedge x \not F y) \vee (x F y \wedge \min(PATH_{x,[y]_E}) \text{ ends in } z) \right)$$

Before expanding the last part it is useful introducing the following notation: given $\vec{l} \in \omega^{<\omega}$, define

$$\langle x, g_{\vec{l}} \cdot x \rangle := \begin{cases} \langle x \rangle & \text{if } \text{length}(\vec{l}) = 0 \\ \langle x, g_{\vec{l} \upharpoonright \text{length}(\vec{l})-1} \cdot x \rangle \smallfrown \langle (\prod_{i \leq \text{length}(\vec{l})} g_{\text{length}(\vec{l})-i}) \cdot x \rangle & \text{else} \end{cases}$$

Now $\min(PATH_{x,[y]_E})$ ends in z can be expressed as:

$$\begin{aligned} \exists n < \omega \exists \vec{i} \in \omega^n & \left((\langle x, g_{\vec{i}} \rangle \text{ is } T\text{-path ending in } z) \wedge \right. \\ & (\forall k < n \forall \vec{j} \in \omega^k \langle x, g_{\vec{j}} \cdot x \rangle(k) \in [y]_E \rightarrow \langle x, g_{\vec{j}} \cdot x \rangle \text{ is not } T\text{-path}) \wedge \\ & \left. (\forall \vec{l} \in \omega^n \langle x, g_{\vec{j}} \cdot x \rangle(k) \in [y]_E \rightarrow \langle x, g_{\vec{i}} \cdot x \rangle \leq_{x,[y]_E} \langle x, g_{\vec{l}} \cdot x \rangle) \right) \end{aligned}$$

- t does not contain cycles

By contradiction, let $\langle r_0 = s_0, r_1, \dots, r_n, r_{n+1}, s_{k+1}, \dots, s_1, s_0 = r_0 \rangle$ be a t -cycle of minimal length. Let $a_{2i-1} T a_{2i}$ witness $r_{i-1} t r_i$. Let $b_{2i-1} T b_{2i}$ witness $s_{i-1} t s_i$. Since the t -cycle is minimal, the elements are distinct. Therefore the same holds for a_i, b_j (since ρ is a function). Consider the T -path passing through $r_0, a_1, a_2, r_1, \dots, a_{2n+2}, r_{n+1} = s_{k+1}, \dots, b_1, s_0$. This is a T -cycle, contradiction.

- $\{\text{classes of } E\} = \{\text{connected components of } t\}$

Denote with $[x]_t$ the connected component of x . It is enough proving that $\forall x \in X ([x]_E = [x]_t)$.

$$- [x]_t \subseteq [x]_E :$$

$$\begin{aligned} y \in [x]_t &\Rightarrow \exists z_1, \dots, z_k (x t z_1 t \dots t z_k t y) \\ &\Rightarrow x E z_1 E \dots E z_k E y \\ &\Rightarrow x E y \end{aligned}$$

$$- [x]_E \subseteq [x]_t :$$

$$\begin{aligned} y \in [x]_E &\Rightarrow y E x \\ &\Rightarrow y F x \\ &\Rightarrow \exists z_1, \dots, z_k (y =: z_0 T z_1 T \dots T z_k T z_{k+1} := x) \\ &\Rightarrow \bigwedge_{i=0}^k \left(\rho(z_i, x) = \rho(z_{i+1}, x) \vee \rho(z_i, x) t \rho(z_{i+1}, x) \right) \end{aligned}$$

At this point, deleting the duplicates (if any) in $\langle y = \rho(z_0, x), \dots, \rho(z_{k+1}, x) = x \rangle$, one obtains the desired t -sequence witnessing $y \in [x]_E$.

□

Esercizio 4. Let X, Y be standard Borel spaces, $a : G \curvearrowright X, b : G \curvearrowright Y$ be Borel actions on X, Y respectively. Suppose that $f : X \rightarrow Y$ be orbit-wise injective, G equivariant, Borel map. Prove that if E_b has the cocycle property, then so does E_a .

Proof. Let $\rho : E_b \rightarrow G$ be a cocycle. Consider $\delta : E_a \rightarrow G, (x_1, x_2) \mapsto \rho(f(x_1), f(x_2))$. We need to prove the following claims.

- δ is well defined:

We need to verify that $\forall (x_1, x_2) \in E_a \left((f(x_1), f(x_2)) \in E_b \right)$. To see this observe that $(x_1, x_2) \in E_a \Rightarrow x_1 = g \cdot x_2 \Rightarrow f(x_1) = f(g \cdot x_2) = g \cdot f(x_2) \Rightarrow f(x_1) E_b f(x_2)$

- δ is cocycle for E_a :

We need to verify the following:

- $\delta(x_1, x_3) = \delta(x_2, x_3) \cdot \delta(x_1, x_2)$:
 $\delta(x_2, x_3) \cdot \delta(x_1, x_2) = \rho(f(x_2), f(x_3)) \cdot \rho(f(x_1), f(x_2)) = \rho(f(x_1), f(x_3)) = \delta(x_1, x_3)$
- $\delta(x_1, x_2) \cdot x_1 = x_2$:
First observe that $x_1 E_a x_2 \Rightarrow \delta(x_1, x_2) \cdot x_1 E_a x_2$. Now, by orbit-wise injectivity of f we just need to check that $f(\delta(x_1, x_2) \cdot x_1) = f(x_2)$. This is true because $f(\delta(x_1, x_2) \cdot x_1) = \delta(x_1, x_2) \cdot f(x_1) = \rho(f(x_1), f(x_2)) \cdot f(x_1) = f(x_2)$

□